

Objective

With the rise of large-scale data systems, cyberattacks have become a major security concern. Manual monitoring of network activity is inefficient and error-prone. This project develops a machine learning-based threat detection system that automates the identification and classification of cyber threats.

Trained on a dataset of 2.4 million network flows with 116 features, the model detects 14 attack types, including DoS, DDoS, Port Scans, Botnet activity, and web intrusions. By comparing multiple ML algorithms, the project identifies the most effective approach for real-time, accurate, and scalable intrusion detection.

Impact:

This work shows how machine learning strengthens cybersecurity by reducing detection time, minimizing false positives, and enabling proactive defense against evolving threats.

Methodology

Dataset: 2,438,052 network flows with 116 traffic features.
<https://www.kaggle.com/network-intrusion-detection/data>

Pre-Processing:

- Removed irrelevant features (IPs, timestamps, ports)
- Handled missing values with median imputation
- Removed constant features (3 features)
- Applied StandardScaler normalization

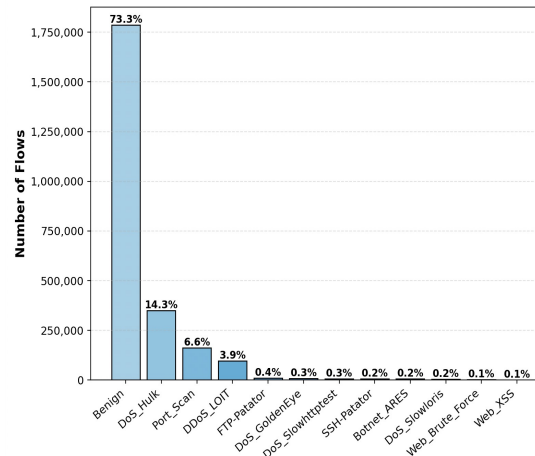
Train/Test Split: 80/20 stratified split (1.95M train, 487K test)

Models Evaluated: Random Forest, Decision Tree, Naive Bayes. Focused on accuracy, scalability, and detection efficiency.

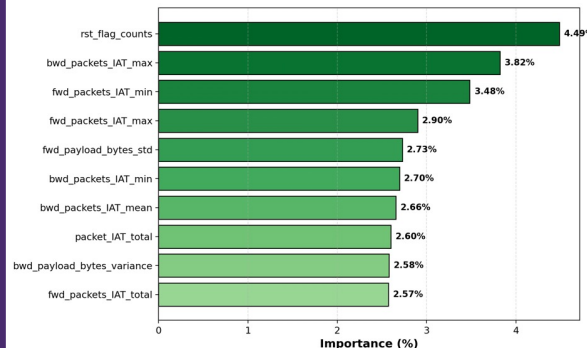
Threat Categories: 14 types (Benign, DoS variants, DDoS, Port Scan, FTP/SSH-Patator, Botnet, Web attacks, and Heartbleed).

Analysis

Threat Distribution in Dataset
(2,438,052 Total Flows)



Top 10 Feature Importance
(Random Forest Model)



Results and Findings

Model Accuracy:

- Random Forest: 99.74% - Best Performer
- Decision Tree: 99.72%
- Naive Bayes: 53.17%

Key Findings:

- Random Forest achieved near-perfect detection across all 14 threat categories.
- Exhibited 100% precision and recall for major threats such as DoS_Hulk, Port_Scan, and DDoS_LOIT.
- Minor performance limitations observed with rare classes such as Heartbleed, which only has 2 samples.
- Top contributing features: RST flag counts, packet timing metrics, and payload-based statistics.

Conclusion and Future Work

The **Random Forest model** performed very well for multi-class threat detection, reaching **99.74% accuracy**. It was able to detect both common attacks like DoS and DDoS and more complex threats such as Botnet and web attacks.

Limitations:

- Rare attacks like Heartbleed were hard to detect due to very few samples.
- Web-based attacks had slightly lower precision (71–79%).

Future Work:

- Deploy the model for real-time network monitoring.
- Use deep learning to improve detection of rare threats.
- Add explainability features to help analysts understand the model's decisions.

References

- Brownlee, J. (2016). *Machine Learning Mastery with Python*. [PDF]. Available at GitHub.
- YouTube. ML Model Tutorials. Available at: <https://youtu.be/Hr06nSA-qww>
- GitHub Repo: <https://github.com/Sai-Lakku/ML-ID.git>

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